PROJECT SYNOPSIS

Title: Explainable Machine Learning Framework for Offline Signature Forgery Detection

INTRODUCTION:

Offline signature verification and forgery detection play a crucial role in secure identity authentication across domains such as banking, legal documentation, and administrative systems. With the rapid advancement of forgery techniques, traditional signature-based authentication methods often fail to provide reliable results. This creates a strong need for intelligent systems that can both verify signatures and detect forged samples accurately. Machine learning-based approaches have shown promise in addressing these challenges. However, the lack of transparency in existing systems limits their trustworthiness in real-world usage.

MOTIVATION BEHIND THE PROJECT WORK:

The motivation behind this project arises from the limitations of existing signature verification and forgery detection systems that focus mainly on accuracy. Most current approaches operate as black-box models, making it difficult to understand how decisions are made. This lack of interpretability reduces trust, especially in sensitive applications. Our project aims to bridge this gap by introducing explainability into the machine learning framework. By doing so, the system not only detects forged signatures but also clearly explains the reasoning behind each decision

PROBLEM STATEMENT DEFINITION:

Despite advancements in machine learning techniques for offline signature verification and forgery detection, existing systems lack transparency and interpretability. These systems provide a final decision without explaining the contributing factors, which is a major drawback in forensic and validation-based applications. There is a need for a framework that can accurately verify signatures, detect forgeries, and also justify its decisions. Addressing this problem is essential to improve trust, accountability, and usability. This project focuses on solving these challenges through an explainable machine learning approach**.**

STATEMENT OF THE OBJECTIVES:

- To design a machine learning-based framework for offline signature verification and forgery detection.
- To accurately classify signatures as genuine or forged using extracted features.
- To introduce explainability into the decision-making process to improve transparency.
- To enhance user trust by providing understandable reasoning for each classification outcome.
- To develop a reliable and interpretable system suitable for real-world applications.

POSSIBLE OUTPUT OF THE PROJECT:

The project will result in an effective offline signature verification and forgery detection system with improved transparency. Users will be able to understand not only the classification result but also the reasoning behind it. The system will achieve a balance between accuracy and interpretability. This approach will increase confidence in automated signature analysis systems. The project will also demonstrate the importance of explainable machine learning in security-related applications.